In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use("ggplot")
%matplotlib inline
```

In [2]:
```
import os

print("Current Working Directory:")
print(os.getcwd())
```

```
Current Working Directory:
c:\Users\palak\Desktop\Daily_Tasks\TASK5_EDA_Financial_Data_Analysis\notebook
```

In [3]:
```
import os

print("Current Working Directory:")
print(os.getcwd())

print("\nFiles and folders here:")
print(os.listdir())
```

```
Current Working Directory:
c:\Users\palak\Desktop\Daily_Tasks\TASK5_EDA_Financial_Data_Analysis\notebook

Files and folders here:
['EDA.ipynb']
```

In [4]:
```
import os

for root, dirs, files in os.walk("."):
    for file in files:
        if file.endswith(".csv"):
            print(os.path.join(root, file))
```

In [5]:
```
df = pd.read_csv("../data/financial_sample.csv")
```

In [6]:
```
df.head()
```

Out[6]:

	Segment	Country	Product	Discount Band	Units Sold	Manufacturing Price	Sale Price	Gross Sales
0	Government	Canada	Carretera	None	1618.5	$3.00	$20.00	$32,370.00
1	Government	Germany	Carretera	None	1321.0	$3.00	$20.00	$26,420.00
2	Midmarket	France	Carretera	None	2178.0	$3.00	$15.00	$32,670.00
3	Midmarket	Germany	Carretera	None	888.0	$3.00	$15.00	$13,320.00
4	Midmarket	Mexico	Carretera	None	2470.0	$3.00	$15.00	$37,050.00

In [7]:
```
df.shape
```

Out[7]: (700, 16)

In [8]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 700 entries, 0 to 699
Data columns (total 16 columns):
 #   Column                 Non-Null Count  Dtype  
---  ------                 --------------  -----  
 0   Segment                700 non-null    object 
 1   Country                700 non-null    object 
 2    Product               700 non-null    object 
 3    Discount Band         700 non-null    object 
 4   Units Sold             700 non-null    float64
 5    Manufacturing Price   700 non-null    object 
 6    Sale Price            700 non-null    object 
 7    Gross Sales           700 non-null    object 
 8    Discounts             700 non-null    object 
 9     Sales                700 non-null    object 
 10   COGS                  700 non-null    object 
 11   Profit                700 non-null    object 
 12  Date                   700 non-null    object 
 13  Month Number           700 non-null    int64  
 14   Month Name            700 non-null    object 
 15  Year                   700 non-null    int64  
dtypes: float64(1), int64(2), object(13)
memory usage: 87.6+ KB
```

In [9]:
```
df.describe(include="all")
```

Out[9]:

	Segment	Country	Product	Discount Band	Units Sold	Manufacturing Price	Sale Price
count	700	700	700	700	700.000000	700	700
unique	5	5	6	4	NaN	6	7
top	Government	Canada	Paseo	High	NaN	$10.00	$20.00
freq	300	140	202	245	NaN	202	100
mean	NaN	NaN	NaN	NaN	1608.294286	NaN	NaN
std	NaN	NaN	NaN	NaN	867.427859	NaN	NaN
min	NaN	NaN	NaN	NaN	200.000000	NaN	NaN
25%	NaN	NaN	NaN	NaN	905.000000	NaN	NaN
50%	NaN	NaN	NaN	NaN	1542.500000	NaN	NaN
75%	NaN	NaN	NaN	NaN	2229.125000	NaN	NaN
max	NaN	NaN	NaN	NaN	4492.500000	NaN	NaN

In [10]:
```
df.isnull().sum()
```

Out[10]:
```
Segment                  0
Country                  0
 Product                 0
 Discount Band           0
Units Sold               0
 Manufacturing Price     0
 Sale Price              0
 Gross Sales             0
 Discounts               0
  Sales                  0
 COGS                    0
 Profit                  0
Date                     0
Month Number             0
 Month Name              0
Year                     0
dtype: int64
```

In [11]:
```
df["Segment"].value_counts()
```

Out[11]:
```
Segment
Government          300
Midmarket           100
Channel Partners    100
Enterprise          100
Small Business      100
Name: count, dtype: int64
```

In [12]:
```
df["Country"].value_counts()
```

Out[12]:
```
Country
Canada                      140
Germany                     140
France                      140
Mexico                      140
United States of America    140
Name: count, dtype: int64
```

In [13]:
```
df.columns = df.columns.str.strip()
df.columns
```

Out[13]:
```
Index(['Segment', 'Country', 'Product', 'Discount Band', 'Units Sold',
       'Manufacturing Price', 'Sale Price', 'Gross Sales', 'Discounts',
       'Sales', 'COGS', 'Profit', 'Date', 'Month Number', 'Month Name',
       'Year'],
      dtype='object')
```

In [14]:
```
df["Product"].value_counts()
```

Out[14]:
```
Product
Paseo         202
Velo          109
VTT           109
Amarilla       94
Montana        93
Carretera      93
Name: count, dtype: int64
```

In [15]:
```
currency_cols = [
    "Manufacturing Price",
    "Sale Price",
    "Gross Sales",
    "Discounts",
    "Sales",
    "COGS",
    "Profit"
]

for col in currency_cols:
    df[col] = (
        df[col]
        .astype(str)
        .str.replace("$", "", regex=False)
        .str.replace(",", "", regex=False)
        .str.replace("(", "-", regex=False)
        .str.replace(")", "", regex=False)
        .str.replace("-", "0", regex=False)
    )

    df[col] = pd.to_numeric(df[col], errors="coerce")

df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 700 entries, 0 to 699
Data columns (total 16 columns):
 #   Column               Non-Null Count  Dtype  
---  ------               --------------  -----  
 0   Segment              700 non-null    object 
 1   Country              700 non-null    object 
 2   Product              700 non-null    object 
 3   Discount Band        700 non-null    object 
 4   Units Sold           700 non-null    float64
 5   Manufacturing Price  700 non-null    float64
 6   Sale Price           700 non-null    float64
 7   Gross Sales          700 non-null    float64
 8   Discounts            700 non-null    float64
 9   Sales                700 non-null    float64
 10  COGS                 700 non-null    float64
 11  Profit               700 non-null    float64
 12  Date                 700 non-null    object 
 13  Month Number         700 non-null    int64  
 14  Month Name           700 non-null    object 
 15  Year                 700 non-null    int64  
dtypes: float64(8), int64(2), object(6)
memory usage: 87.6+ KB
```

In [16]:

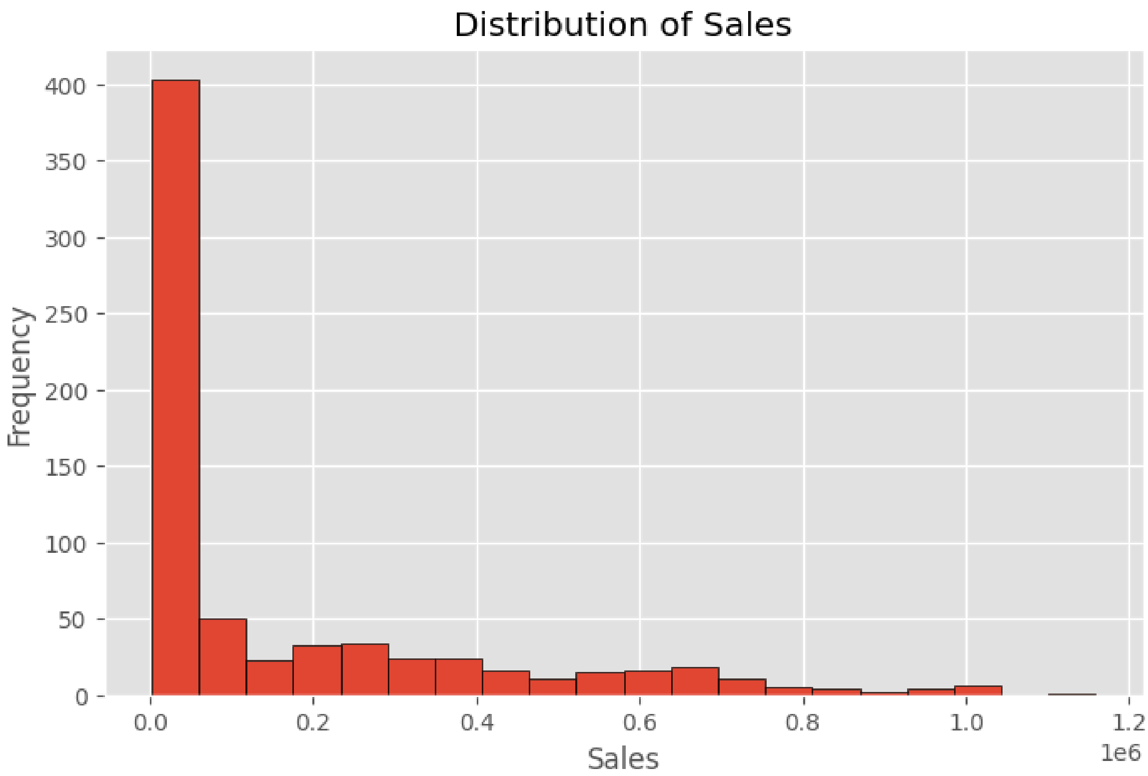
```python
df[~df["Discounts"].astype(str).str.replace("$","",regex=False)
                         .str.replace(",","",regex=False)
                         .str.replace("(","-",regex=False)
                         .str.replace(")","",regex=False)
                         .str.match(r"^-?\d*\.?\d*$", na=False)]["Discounts"].un
```

Out[16]:

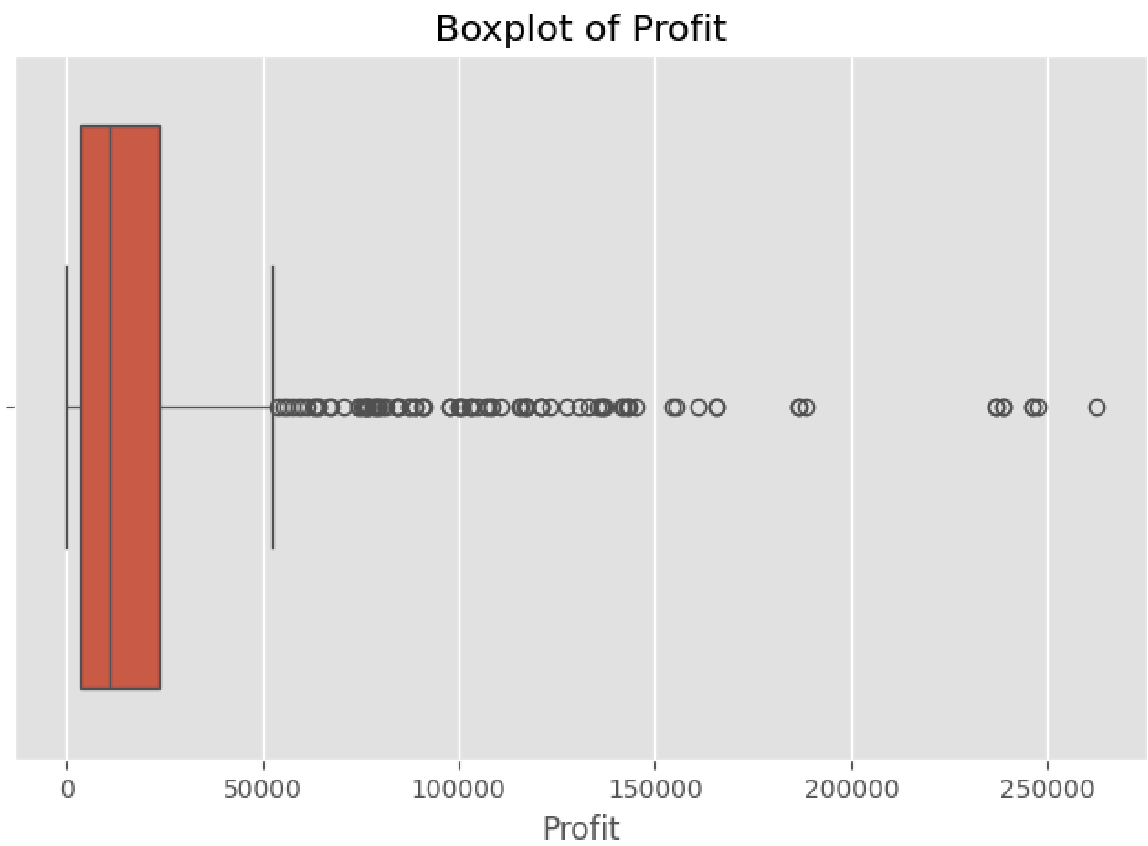
```
array([], dtype=float64)
```

In [17]:

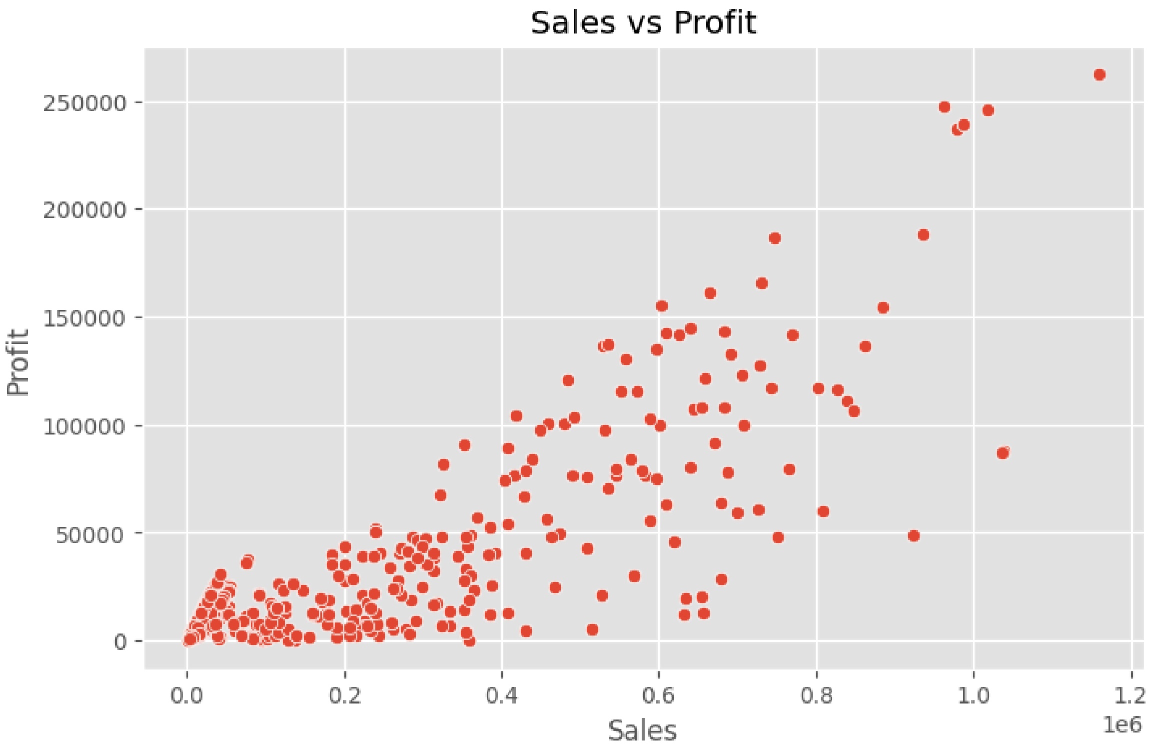
```python
plt.figure(figsize=(8,5))
plt.hist(df["Sales"], bins=20, edgecolor="black")
plt.title("Distribution of Sales")
plt.xlabel("Sales")
plt.ylabel("Frequency")

plt.savefig("../images/histogram_sales.png", dpi=300, bbox_inches="tight")

plt.show()
```



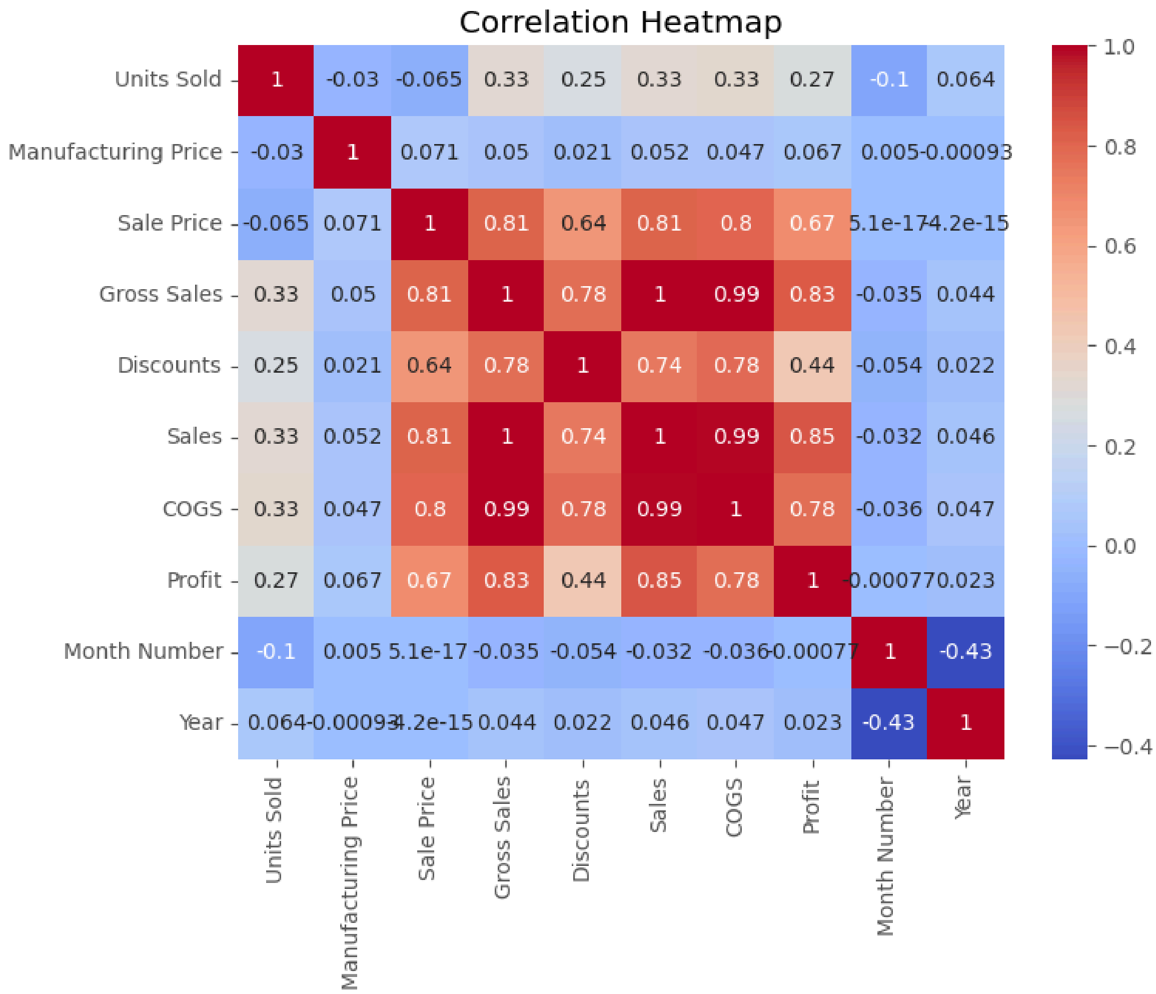
In [18]:
```
plt.figure(figsize=(8,5))
sns.boxplot(x=df["Profit"])
plt.title("Boxplot of Profit")

plt.savefig("../images/boxplot_profit.png", dpi=300, bbox_inches="tight")

plt.show()
```



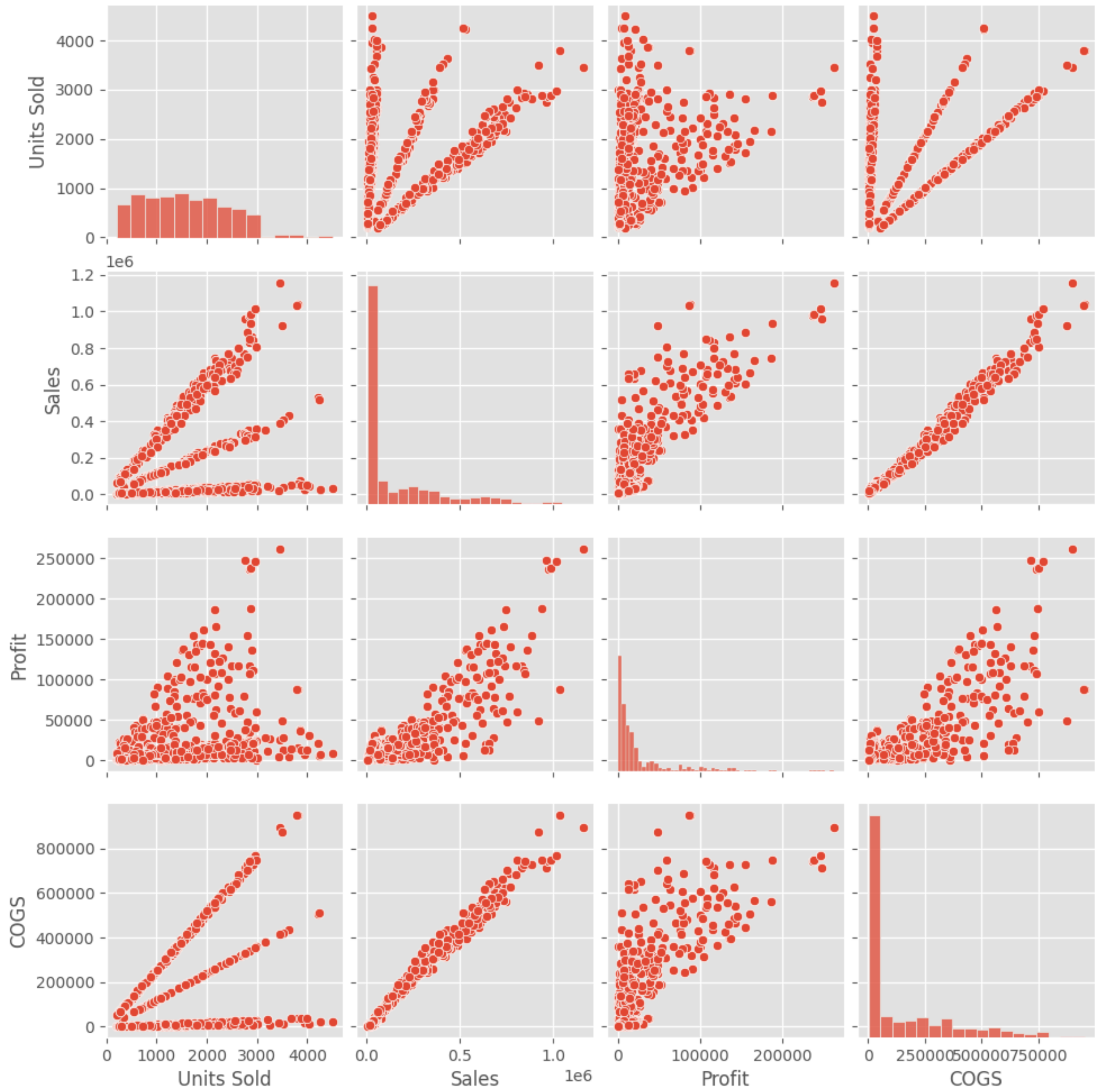
In [19]:
```
plt.figure(figsize=(8,5))
sns.scatterplot(data=df, x="Sales", y="Profit")
plt.title("Sales vs Profit")

plt.savefig("../images/scatter_sales_profit.png", dpi=300, bbox_inches="tight")

plt.show()
```



In [20]:
```
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")

plt.savefig("../images/heatmap.png", dpi=300, bbox_inches="tight")

plt.show()
```




```
In [21]: pair = sns.pairplot(df[["Units Sold", "Sales", "Profit", "COGS"]])

         pair.savefig("../images/pairplot.png", dpi=300, bbox_inches="tight")
```



Summary of Findings

- The dataset contains 700 records and 16 columns.
- No missing values were found.
- Sales distribution is right-skewed with a few high-value transactions.
- Profit contains several outliers.
- Sales and Profit have a strong positive relationship.
- Gross Sales, Sales, COGS, and Profit are highly correlated.
- The dataset is clean and suitable for further predictive analysis.